

Topological Dark Matter and Tully-Fisher Slope in SDGFT

David A. Besemer

Independent Researcher

david.besemer@sdgft.org

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Abstract. We identify dark matter as unclosed topological defects in the 24-cell lattice. We derive the cold dark matter density $\Omega_c = 600/2304 \approx 0.26043$ from the lattice flatness constraint. The Tully-Fisher slope $b_{TF} = 91/24 \approx 3.7917$, matching the observed velocity-mass scaling in galaxies.

1 Introduction

Dark matter and dark energy are usually treated as separate fluids. In SDGFT, they both arise from the topological structure of the 24-cell lattice.

2 Theoretical Formulation

Here we formulate the core mathematical relations governing the system:

$$\Omega_c = \frac{600}{2304} \approx 0.26043, \quad \Omega_b = \frac{115}{2304} \approx 0.04991 \quad (1)$$

$$b_{TF} = \frac{91}{24} \approx 3.7917 \quad (\text{Tully-Fisher exponent}) \quad (2)$$

3 The Tully-Fisher Relation

We show that the Tully-Fisher slope b_{TF} is a direct consequence of the modified gravity potential at galactic scales,

4 Conclusion

Galactic rotation curves are explained without exotic dark matter particles, using the modified gravity potential of the 24-cell lattice.

References

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